

Recall from last class:

- Define $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(\vec{x}) = A\vec{x}$ where A is an m by n matrix. Recall we call $\mathbb{R}^n$ the domain and $\mathbb{R}^m$ the codomain. The set of outputs, $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$, is the image or the range of T .
- $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if
 1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in $\mathbb{R}^n$,
 2. $T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in $\mathbb{R}^n$ and c in $\mathbb{R}$.
- **Properties of Linear Transformations.**
 1. If T is linear, then $T(\vec{0}) = \vec{0}$.
 2. If T is linear, then $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$.
- All matrix transformations are linear.
- If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is given by $T(\vec{x}) = A\vec{x}$, then the image of T is the span of the columns of A ; namely,
$$\text{Im}(T) = \text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}.$$
- We also saw various geometric transformations, including projections, dilations, and shears.

1. Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is linear and

$$T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix}, \text{ and } T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}.$$

What is

$$T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right)?$$

Theorem 10: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then there exists a unique matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.

2. Prove the Theorem.

3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation given by rotating $\vec{x}$ by θ counterclockwise. Find the associated matrix of T .

4. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be defined by $T(\vec{x}) = A\vec{x}$ where

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Is T injective and/or surjective?

Theorem 11: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Then T is injective if and only if $T(\vec{x}) = \vec{0}$ has only the trivial solution.

5. Prove the theorem.

Theorem 12: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear, and let A be the associated matrix. Then

(a) T is surjective if and only if the columns of A span $\mathbb{R}^m$.

(b) T is injective if and only if the columns of A are linearly independent.

6. Prove the theorem.

Linear and Matrix Transformations – Answers / Solutions

1. Using linearity, we get

$$\begin{aligned} T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) &= T\left(a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= aT\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + bT\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}. \end{aligned}$$

Thus T is also a matrix transformation. A natural question is whether every linear transformation is a matrix transformation.

2. We proceed as in the last example. First Let $\vec{e}_j$ be the vector in $\mathbb{R}^n$ with 1 in the j -th position and 0 elsewhere. Let $x \in \mathbb{R}^n$, then

$$\vec{x} = x_1\vec{e}_1 + \cdots + x_n\vec{e}_n.$$

Thus using the linearity of T , we get

$$\begin{aligned} T(\vec{x}) &= x_1T(\vec{e}_1) + \cdots + x_nT(\vec{e}_n) \\ &= [T(\vec{e}_1) \ \cdots \ T(\vec{e}_n)] \vec{x}. \end{aligned}$$

Thus $T(\vec{x}) = A\vec{x}$ where $A = [T(\vec{e}_1) \ \cdots \ T(\vec{e}_n)]$.

In order to prove uniqueness, we suppose there is another matrix B such that $T(\vec{x}) = B\vec{x}$ for all $x \in \mathbb{R}^n$. We want to show that $A = B$. Let $B = [\vec{b}_1 \ \cdots \ \vec{b}_n]$. Since $A\vec{e}_j = B\vec{e}_j$, we get $\vec{b}_j = T(\vec{e}_j)$. Thus A and B have the same columns, and thus $A = B$ and A is unique.

3. Rotating $\vec{e}_1$ and $\vec{e}_2$ by θ gives the associated matrix of T is $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.
4. We have a pivot position in each row, so T is onto (surjective). $A\vec{x} = \vec{0}$ has free variables, so T is not one-to-one (injective).
5. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear.

$\Rightarrow$ Suppose T is one-to-one (injective). Since T is linear, we know $\vec{x} = \vec{0}$ is a solution. Since T is injective, we know $T(\vec{x}) = \vec{0}$ has at most one solution. Thus $T(\vec{x}) = \vec{0}$ has only the trivial solution.

$\Leftarrow$ Suppose $T(\vec{x}) = \vec{0}$ has only the trivial solution and $T(\vec{u}) = T(\vec{v})$. Since T is linear, we get $T(\vec{u} - \vec{v}) = \vec{0}$. Since $T(\vec{x}) = \vec{0}$ has only the trivial solution, we know that $\vec{u} - \vec{v} = \vec{0}$, and hence $\vec{u} = \vec{v}$. Therefore T is one-to-one or injective.

6. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and let A be the associated matrix.

Proof of (a): T is surjective means that for all $\vec{b} \in \mathbb{R}^m$ there exists $\vec{x} \in \mathbb{R}^n$, or equivalently $A\vec{x} = \vec{b}$ is consistent. By Theorem 4, $A\vec{x} = \vec{b}$ is consistent for all $\vec{b}$ if and only if the columns of A span $\mathbb{R}^m$.

Proof of (b): By Theorem 11, T is one-to-one if and only if $T(\vec{x}) = \vec{0}$ has only the trivial solution. Thus the columns of A must be linearly independent.